

Diagnostics-Infrastructure Retention Tension

Diagnostics needs state retained ↔ Infrastructure needs state discarded



Diagnostics

Infrastructure

1. Retention-aware eviction

Table 6: Eviction criterion

Serving policy pins state flagged as suspect by online detector



Bounded memory cost → preserved replay evidence

2. Failure-aware scheduling

Table 6: Overhead budget

Attribution signals fed back into request placement



Known-unreliable paths deprioritised

3. Budgeted observability

Table 6: Granularity

Trace granularity negotiated against serving SLO at runtime



Dynamic fidelity instead of fixed design-time choice

4. Infrastructure-triggered diagnostics

Table 6: Runtime state

Serving anomalies (latency spikes, memory pressure) as early warnings



Diagnosis begins before task-level failure surfaces

All four are proposals grounded in verified capabilities of existing components; none has been implemented - the measurement our field currently lacks.